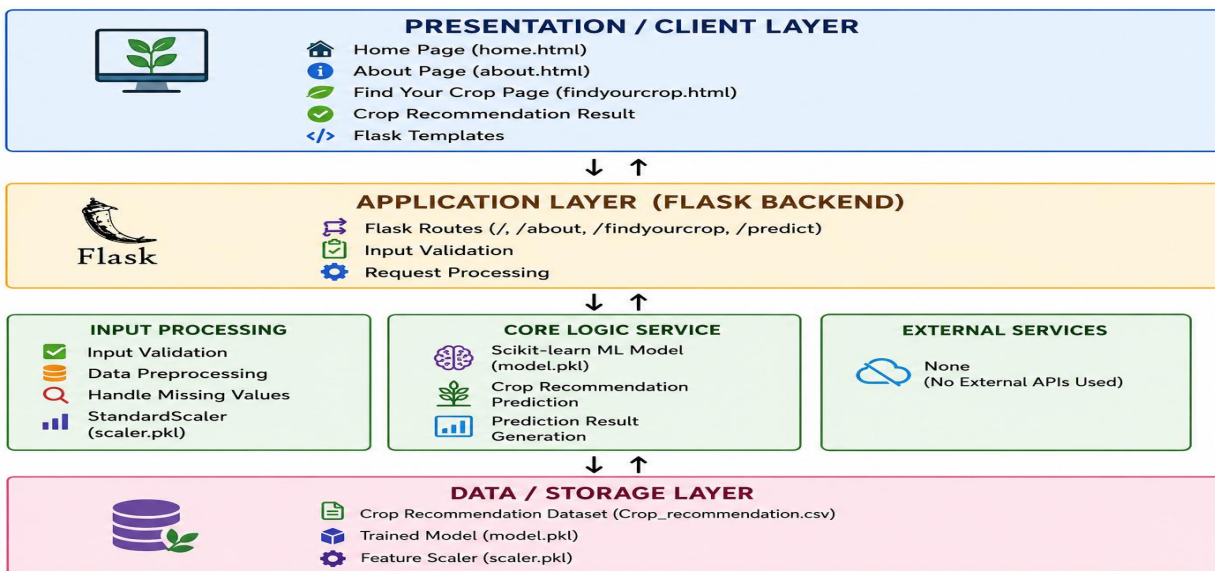


Solution Architecture

Date	27 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI-Based Crop Recommendation System.
Maximum Marks	5 Marks

Solution Architecture Diagram:

The **OptiCrop** solution follows a **three-tier architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. The Presentation Layer provides a user-friendly web interface where users enter soil nutrient values and environmental conditions. The Flask backend receives the input, validates and preprocesses the data using the trained scaler, and communicates with the Machine Learning model to generate the most suitable crop recommendation predictions.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects soil nutrient and environmental parameters from users and displays crop recommendation results through web pages.	HTML, CSS, Flask Templates
Application Layer (Flask Backend)	Handles routing, input validation, request processing, and communicates with the trained Machine Learning model.	Python, Flask
Core Logic Service	Preprocesses user input using StandardScaler and predicts the most suitable crop using the trained Machine Learning model.	Scikit-learn, Joblib/Pickle
Data / Storage Layer	Stores the Crop Recommendation dataset, trained Machine Learning model, and feature scaler required for prediction.	CSV Dataset, model.pkl, scaler.pkl